

FREEDOM INTERNATIONAL SCHOOL

WORKSHEET-MCQ (SOLUTIONS)

PHYSICS

CLASS XII

ALTERNATING CURRENT

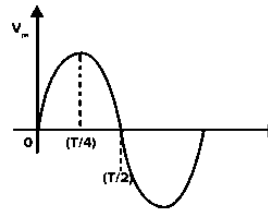
1. An a.c source is rated at 220 V, 50 Hz. The time taken for voltage to change from its peak value to zero is

(a) 50 s (b) 0.02 s (c) 5 s (d) 5×10^{-3} s

Ans: (d) 5×10^{-3} s

$$T = 1/50 \text{ s}$$

Required time for ac to reach it's peak value is $t = \frac{T}{4}$

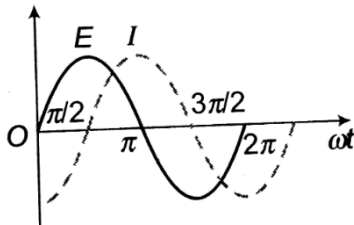


2. In an a.c series circuit, the instantaneous current is maximum when the instantaneous voltage is maximum. The circuit element connected to the source will be

(a) pure inductor (b) pure capacitor
(c) pure resistor (d) combination of a capacitor and an inductor

Ans: (c) pure resistor

3. The variation of instantaneous current $I(t)$ and instantaneous emf $\varepsilon(t)$ versus ωt for a circuit is shown below.

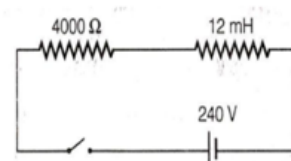


Identify the correct statement

(a) the voltage lags behind the current by $\pi/2$ (b) the voltage leads the current by $\pi/2$
(c) the voltage and the current are in phase (d) the voltage leads the current by π

Ans: (b) the voltage leads the current by $\pi/2$

4. In the inductive circuit given in the figure, the current rises after the switch is closed. At instant when the current is 15 mA, then potential difference across the inductor will be



(a) zero (b) 240 V (c) 180 V (d) 60 V

Ans: (c) 180 V

$$I = 15 \times 10^{-3} \text{ A}, R = 4000 \Omega$$

$$V_R + V_L = 240 \text{ V}$$

$$V_L = 240 - IR = 180 \text{ V}$$

5. In a step-up transformer, 220 V is converted into 2200 V. The number of turns in primary coil is 600. What is the number of turns in the secondary coil?

- (a) 60 (b) 600 (c) 6000 (d) 100

Ans: (c) 6000

$$V_p = 220 \text{ V}; V_s = 2200 \text{ V}; N_p = 600$$

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

$$N_s = (2200/220)(600)$$

6. In a series LR- circuit $X_L = 3 R$. Now a capacitor with $X_C = R$ is added in series. Ratio of new to old power factor is

- (a) $\sqrt{2}$ (b) $1/\sqrt{2}$ (c) 2 (d) 1

Ans: $\sqrt{2}$

In series LR circuit,

$$\text{Power factor, } \cos \phi_1 = \frac{R}{Z}$$

$$= \frac{R}{\sqrt{R^2 + X_L^2}} = \frac{R}{\sqrt{R^2 + 9R^2}}$$

$$= \frac{R}{R\sqrt{10}} = \frac{1}{\sqrt{10}}$$

When capacitor with $X_C = R$ is added in series, new power factor,

$$\phi_2 = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}}$$

$$\cos \phi_2 = \frac{R}{\sqrt{R^2 + (3R - R)^2}} = \frac{R}{R\sqrt{5}}$$

$$\therefore \frac{\cos \phi_2}{\cos \phi_1} = \frac{1}{\sqrt{5}} \frac{\sqrt{10}}{1} = \sqrt{2}$$

7. The power delivered by the source of circuit is maximum when

- (a) $\omega L = \omega C$ (b) $\omega L = 1/\omega C$ (c) $\omega L = \omega C^2$ (d) $\omega L = \sqrt{\omega C}$

Ans: (b) $\omega L = 1/\omega C$

8. In an AC circuit, V and I is given by

$$V = 1000 \sin(1000 t) \text{ volt}$$

$$I = 100 \sin(100 t + \frac{\pi}{6}) \text{ mA}$$

The power dissipation in the circuit in one complete cycle is

- (a) 25 W (b) $25\sqrt{3}$ W (c) 100 W (d) 10 W

Ans: (b) $25\sqrt{3}$ W

Power dissipation in an AC circuit in one complete cycle is given by ,

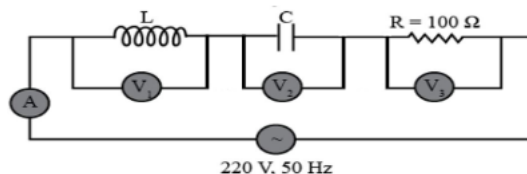
$$P_{av} = (1/2)V_o I_o \cos \phi$$

given , $V_o = 1000V$, $I_o = 100mA = 0.1A$, $\phi = \pi/6 = 30^\circ$

Hence , $P_{av} = (1/2)1000 \times 0.1 \times \cos 30 = 50 \times (\sqrt{3}/2)$

or $P_{av} = 25\sqrt{3} \text{ W}$

9. In the given circuit the readings of voltmeter V_1 and V_2 are 300 volts each. The readings of the voltmeter V_3 and ammeter A are respectively



- (a) 150 V, 2.2 A (b) 220 V, 2.2 A (c) 220 V, 2.0 A (d) 100 V, 2.0 A

Ans: (b) 220 V, 2.2 A

As $V_L = V_C = 300 \text{ V}$,

therefore the given series LCR circuit is in resonance.

$$\therefore V_R = V = 220 \text{ V}, Z = R = 100 \Omega$$

$$\text{Current, } I = \frac{V}{Z} = \frac{220 \text{ V}}{100 \Omega} = 2.2 \text{ A}$$

10. The magnetic potential energy stored in a certain inductor is 25 mJ, when the current in the inductor is 60 mA. This inductor is of inductance

- (a) 0.138 H (b) 13.89 H (c) 1.389 H (d) 138.88 H

Ans: (b) 13.89 H

$$U = 25 \times 10^{-3} \text{ J}; I = 60 \times 10^{-3} \text{ A}$$

$$U = \frac{1}{2} LI^2$$

$$L = \frac{2U}{I^2} = 2 \times 25 \times 10^{-3} / (3600 \times 10^{-6})$$

For questions 11 to 15, two statements are given- one labelled Assertion (A) and other labelled Reason (R). Select the correct answer to these questions from the options as given below.

- A. Both Assertion and Reason are true and Reason is the correct explanation of Assertion.**
B. Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
C. Assertion is true but Reason is false.
D. Both Assertion and Reason are false.

11. **Assertion:** The capacitive reactance limits the amplitude of the current in a purely capacitive circuit.

Reason: capacitive reactance is proportional to the frequency and capacitance.

Ans: C

12. **Assertion:** In a purely inductive or capacitive circuit, the current is referred to as wattless current.

Reason: No power is dissipated in a purely inductive or capacitive circuit, even though a current is flowing in the circuit.

Ans: A

13. **Assertion:** When a current flows in the coil of a transformer then its core becomes hot.

Reason: The core of transformer is made of soft iron.

Ans: B

14. **Assertion:** When ac circuit contain resistor only, its power is minimum.

Reason: Power of a circuit is independent of phase angle.

Ans: D

15. **Assertion:** A laminated core is used in transformers to increase eddy currents.

Reason: The efficiency of a transformer increases with increase in eddy currents.

Ans: D